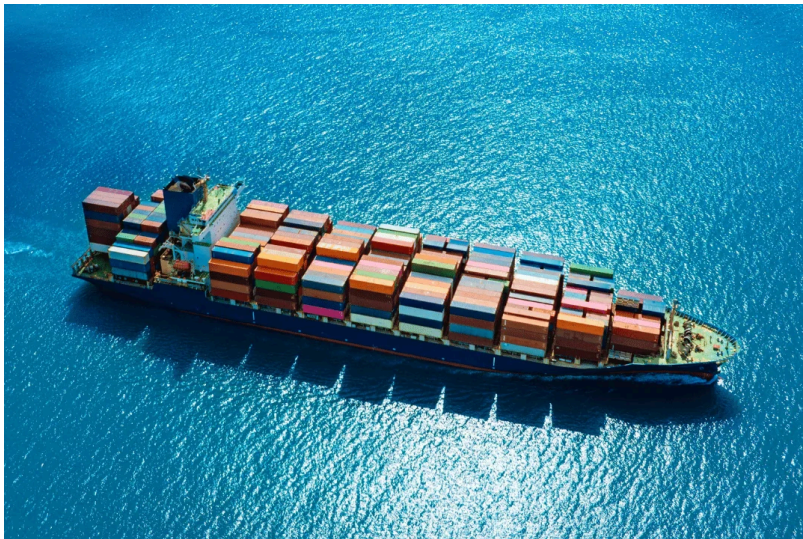




Capstone Project

Advanced Optimization and Simulation

Shanghai-New York Container Shipping Reliability: A Monte Carlo Simulation Analysis



20th April 2026

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1. Introduction

1.1 Background and the Reliability Metric

Container shipping is one of the most significant contributors to international trade, moving approximately 11 billion tonnes of cargo annually. According to the United Nations Conference on Trade and Development (UNCTAD, 2024), around 80 percent of the volume of international trade in goods is carried by sea, a percentage that is even higher for most developing countries (World Bank, 2024). In the maritime industry, there are several ways to look at what reliability actually implies: network reliability, connectivity reliability, infrastructure reliability, and schedule reliability (Yue & Mangan, 2024). While all are important, schedule reliability is the most commonly used metric. It refers to the percentage of ships that actually arrive at the port on their specifically scheduled day.

1.2 The 2025 Reliability Crisis

Recently, there has been a sharp decline in these global schedule reliability for maritime transport. During the February to April 2025 period, Beacon Maritime reported that global schedule reliability collapsed to just 6.7 percent (Beacon / Seatrade Maritime, 2026). This represents a massive drop from the 36 percent on-time performance recorded in 2024. Furthermore, 63.9 percent of shipments experienced severe delays exceeding three days, showing that late arrivals have become a systemic issue rather than just a rare occurrence (Beacon / Seatrade Maritime, 2026).

1.3 Impact of Delays and the Suez Canal Example

These delays are associated with higher costs and create trickle down effects across the global supply chain. A clear example of how these delays create bottlenecks is the 2021 Suez Canal blockage, where the Ever Given container ship blocked the canal and caused congestion for a week (Aon, 2021). Every day of that delay held up USD 9 billion of global trade and increased financial pressure at every level of the supply chain (Aon, 2021). Beyond the immediate blockage, the incident also triggered a massive surge in container freight rates, and port congestion in European ports up to several weeks later (UNCTAD, 2024; Aon, 2021).

1.4 The Research Gap

Extreme situations like the Suez blockage are rare, but uncertainties like weather, port infrastructure, and political disruptions affect schedule reliability regularly. Despite how important this is for the global economy, maritime reliability is researched much less than other modes like rail, road, or air. For instance, recent systematic reviews indicate that only 4.2 percent of publications on network reliability between 1950 and 2020 were related to the maritime mode

(Zhang et al., 2024). This lack of research, combined with the current decline in reliability metrics, is exactly why I decided to simulate these journeys.

2. Problem Description

The primary challenge for logistics companies and businesses that directly receive their goods from the port is the difference between the carrier schedule and the actual time the containers leave the port. When a business receives a quote from a carrier for a specific route, they build their entire downstream supply chain and inventory plans around that specific date. However, if a shipper accepts this quote without verifying how realistic it is, they risk a situation where small delays in the early stages of the journey compound into much larger disruptions at the final destination.

The objective of this study is to provide a method for these businesses to evaluate the reliability of a carrier quote. By accounting for necessary buffer times, businesses can better prepare for potential delays and avoid the costs associated with stockouts and emergency logistics. This approach provides more stability for their own internal systems and helps them maintain a good impression with their own customers by ensuring products are available as promised. This study addresses this problem by simulating a high complexity route across multiple different scenarios to see the true probability of the ships actually being on time.

3. Shanghai to New York Route Selection

I chose the Shanghai to New York corridor as the case study because it represents a high level of operational complexity and is critical for global trade.

- **Economic Volume and Trade Sensitivity:** This corridor is one of the busiest in the world, with the Port of Shanghai consistently ranking as the global leader in container volume and the Port of New York and New Jersey serving as the largest gateway on the US East Coast, handling over 7.8 million TEUs annually (UNCTAD, 2024; Port Authority of NY & NJ, 2025). As a critical maritime trade lane for high-value electronics and pharmaceuticals, any schedule deviation on this route causes immediate financial losses for the receiving business.
- **Operational Bottlenecks:** Unlike a direct Pacific crossing to the West Coast, the route to New York involves the Panama Canal and a transshipment at a hub like Cartagena. These extra steps provide multiple points where small delays can compound.
- **Environmental and Labor Volatility:** This specific corridor is currently facing significant risks, including historic drought related bottlenecks at the Panama Canal and labor tensions at North American ports. These factors make it an ideal environment for testing how different risks affect the total transit time.

4. Rationale for Monte Carlo Simulation

4.1 Ability to Handle Multi-Level Uncertainty

The primary reason I chose Monte Carlo simulation is its ability to manage the multilevel uncertainties inherent in global supply chains. It simultaneously integrates the different probability distributions of every leg of the route to capture how these independent risks compound over thousands of iterations to reveal the total impact on transit and lead times.

4.2 Managing Computational Complexity

Monte Carlo addresses the mathematical complexity of calculating every possible system state by using randomization to ensure all potential scenarios are represented without the need to explicitly cover every single permutation.

5. Model Formulation

5.1 Key Decision Variables and Parameters

The model is built as a sequential system where the total transit time is the summation of the time taken for each of the six individual legs of the journey.

$$\text{Total Transit Time} = \text{Leg 1} + \text{Leg 2} + \text{Leg 3} + \text{Leg 4} + \text{Leg 5} + \text{Leg 6}$$

The key metrics tracked by the model include:

- **Average Vessel Transit Time (Port to Port):** This measures the total time the ship spends moving through the network. It explicitly includes water travel time, canal transits, and transshipment handling. This metric also accounts for essential port operations such as berthing, loading, and unloading at the docks. While these activities typically take anywhere between 1 to 3 days depending on vessel capacity and terminal efficiency, I have standardized these operations to 1.25 days for this model to prevent introducing excessive statistical noise.
- **Average Cargo Lead Time (Total Journey):** This is a comprehensive metric for the business receiving the goods. It includes the full Vessel Transit Time plus the pre-departure and post-arrival port dwell times. This represents the total time from when the cargo is ready at the origin to when it is actually available for pickup at the destination.
- **Average Cargo Availability Gap:** This represents the time lost during port dwell and landside processing, measured as the gap between vessel berthing and the container clearing the port gates.
- **On Time Probability:** This is the percentage of simulated journeys that met or stayed below the carrier express quote of 38 days.

- **Severe Delay Probability:** This measures the percentage of journeys that exceed 45 days.
- **Average Delay:** This measures the mean number of days that shipments exceed the 38 day marketed quote.

5.2 Assumptions and Constraints

- **Independence of Legs:** For the purpose of this simulation, I assume the probability distributions for each leg are independent, though the cumulative time is additive.
- **Fixed Benchmark:** While carriers in the real world may update their schedules weekly or adjust quotes mid transit due to emerging delays, I have assumed a fixed 38 day benchmark to maintain a consistent baseline for measuring reliability.
- **Missed Connection Delay** I have applied a 7 day waiting time if a delay if the first half of the trip causes the ship to miss its departure at the transshipment Cartagena, simulating the reality of waiting a full week for the next available vessel.

5.3 Mathematical Logic: The Triangular Distribution

To account for the uncertainty in each leg, I used the Triangular Distribution, defined by the Minimum (a), Mode (b), and Maximum (c).

The simulation uses the Inverse Cumulative Distribution Function shown below to translate random probabilities into specific daily durations:

$$F^{-1}(p) = \begin{cases} a + \sqrt{(b-a)(c-a)p}, & 0 \leq p < \frac{b-a}{c-a} \\ c - \sqrt{(c-a)(c-b)(1-p)}, & 1 \geq p \geq \frac{b-a}{c-a} \end{cases}$$

To account for different operational conditions, I used two separate triangular distributions assigned with specific probabilities for certain legs. This approach is necessary because maritime disruptions often follow a binary pattern: either the system works efficiently with only basic delays, or a major event/bottleneck occurs that shifts the entire range of possible values. By using two distributions instead of one, the model avoids the mistake of averaging these two extremes into an unrealistic middle value. This ensures the simulation accurately weights both standard operations and rare, severe disruptions.

6. Model Development

6.1 Component Logic

Leg 1: Shanghai Port Dwell

This variable covers the landside waiting and dwell time, representing the period a container spends in the terminal yard after gating in but before it is actually loaded. Activities like vessel berthing and crane operations are excluded here as they are covered in the transit metric.

- **Minimum (0.7 days):** This represents a "just-in-time" scenario where the container arrives at the port just before the vessel's cargo cutoff (CY Cutoff) and gets loaded onto the ship with minimal delay.
- **Mode (1.7 days):** Based on Kuehne + Nagel 2026 operational updates, the most common time is 1.7 days for waiting (Kuehne + Nagel, 2026).
- **Maximum (6.0 days):** Per Linerlytica, a backlog due to fog and Chinese holidays may be up to 6 days long (Linerlytica, 2026).

Leg 2: Pacific Crossing (Shanghai to Panama)

The distance for this leg is roughly 16,000 km. This leg was calculated with the SeaRates Distance and Transit Time Tool (SeaRates, 2026).

Scenario A: Slow Steaming (75% probability)

- Speeds range from 12 to 18 knots according to Neele-Vat Logistics (Neele-Vat, 2026).
- The Mode was taken as 15.6 knots based on the Kuehne + Nagel reference regarding dropping vessel speeds (Kuehne + Nagel, 2026).
- Probability: A ScienceDirect study mentioned 64% of Trans-Pacific ships use slow steaming; I assumed 75% for current modeling to reflect increasing trends towards utilising slow steaming.

Scenario B: Fast Steaming (25% probability)

- For emergency/critical goods like medicines, ships can sail faster from 19 to 24 knots.
- I kept the Mode at 20 knots to keep the range realistic given declining high speed trends.

Leg 3: Panama Canal Transit

I assume a minimum of 1.45 days, including the canal crossing and sailing time to reach Cartagena according to the Panama Canal Authority and SeaRates (Panama Canal Authority, 2026; SeaRates, 2026).

- **Mode (2.75 days):** Based on Lindblad Expeditions, the common time for administrative clearance is 2.75 days (Lindblad, 2026).
- **Maximum (22.0 days):** Per DSV, extreme droughts can cause congestion leading to a severe maximum of 22 days (DSV, 2025).

- **Split Rationale (70-30):** Based on Seatrade Maritime, the canal is recovering to approximately 70% of its total capacity, justifying this split (Seatrade Maritime, 2025).

Leg 4: Cartagena Transshipment Handling

This covers the movement of containers from a Pacific vessel to an Atlantic bound vessel.

- **Minimum (1.10 days):** Covers median wait and crane operations.
- **Mode (2.0 days):** Based on the Kuehne + Nagel average waiting time (Kuehne + Nagel, 2026).
- **Cascade Penalty (+7 days):** If a Pacific vessel misses its connection, a week delay occurs as Atlantic bound ships sail about once a week per SeaRates schedules (SeaRates, 2026).

Leg 5: Atlantic Crossing (Cartagena to New York)

The distance is roughly 3,500 km per SeaRates. The speed ranges and modes (15.6 and 20 knots) remain consistent with the Pacific Crossing leg to maintain uniformity in carrier steaming policies.

Leg 6: New York (NY/NJ) Port Dwell and Landside Waiting

This variable measures the time from when the container is unloaded from the ship at the Port of New York and New Jersey until it is cleared and ready for gate-out.

To accurately model this, I used two distinct scenarios to reflect the volatile nature of US East Coast labor and terminal operations:

- **Normal Operations (80% Probability):** This represents the port's standard functioning environment.
 - **Minimum (1.20 days):** This assumes a fast-track scenario where the port is operating at high efficiency to clear the cargo quickly.
 - **Mode (3.80 days):** Reflects standard terminal processing and typical pick-up windows for the region (Kuehne + Nagel, 2026).
 - **Maximum (7.0 days):** Represents minor seasonal peaks or typical delays.
- **Severe Disruption (20% Probability): This represents events such as labor strikes, contract negotiations, or major equipment failures.**
 - **Maximum (15.0 days):** In cases of severe labor tension or terminal gridlock, clearance can be delayed by up to two weeks (Reuters, 2024).

7. Simulation Analysis

7.1 Testing Methodology and Scale

To ensure the model was statistically sound and not influenced by short term random noise, I conducted an extensive convergence test across several sample sizes. I tested the model at journey counts of 100, 200, 500, 1000, 2500, and 5000. For each journey size, I performed multiple sets of runs (10, 20, 50, 100, and 150) to observe how the mean and standard deviation stabilized.

7.2 Convergence Results and Stability

My initial tests of 100 to 500 journeys proved to be highly volatile, with the mean transit time shifting significantly between runs. However, as the sample size increased, the model started stabilizing. At the final scale of 5000 journeys and 150 runs, the Standard Deviation for Vessel Transit Time was 0.127 days, representing an 85 percent reduction from the initial 0.842 days observed at 100 journeys.

At 5,000 journeys with only 10–20 runs, the Standard Deviation (SD) appeared artificially low (0.107–0.123 days). As the runs increased to 150, the SD stabilized between 0.127 and 0.134 days.

This shift indicates proper sampling coverage rather than instability. Smaller samples can create clusters of similar outcomes, giving a false impression of low variance. Increasing the volume to 150 runs ensures the model captures the true distribution of volatility across independent simulations, providing a statistically reliable estimate of lead-time risk.

8. Results and Analysis

8.1 Statistical Summary (n=5000 Journeys, 150 Runs)

Metric	Mean Value	Standard Deviation (SD)
Vessel Transit Time (Port-to-Port)	40.87 Days	0.127 Days
Cargo Lead Time (Total Journey)	46.78 Days	0.134 Days
Cargo Availability Gap (Port Dwell)	5.91 Days	0.050 Days
Average Delay (Beyond 38-day quote)	2.87 Days	0.127 Days
On-Time Probability (≤ 38 days)	45.98%	0.007%
Severe Delay Probability (≥ 45 days)	28.11%	0.006%

95th Percentile Cargo Lead Time	47.02 Days	
Maximum Observed Transit Time	49.30 Days	

8.2 The 38-Day Baseline vs. Total Lead-Time Volatility

While the carrier's schedule lists a 38-day transit, the simulation reveals that actual delivery dates fall across a vast and volatile range. The mean Cargo Lead Time of 46.78 days exposes a nearly 9-day gap from the schedule, largely driven by the 5.91 days containers spend waiting to clear port gates, a factor often invisible in vessel-only transit data.

Most significantly, the model highlights the risk of extreme outcomes: the 95th percentile stands at 47.02 Days, while the Maximum Lead Time reached 79 days, showing that severe disruptions can more than double the expected duration. With an on-time probability of only 45.98%, meeting the 38-day mark is statistically unlikely. This means that for a shipper, a delay of a week or more is not an uncommon event, but a likely reality that should be considered in inventory and buffer-stock planning.

8.3 28.11% Severe Delay Risk

With nearly one in three containers (28.11%) delayed beyond 45 days, severe disruption is a structural reality rather than a rare outlier. The model demonstrates that when bottlenecks occur at the Panama Canal or US ports, the impact is substantial and non-linear, typically pushing arrivals back by a full week or more. For a shipper, this means nearly 30% of inventory is consistently at risk of being delayed, making it impossible to maintain lean stock levels without significant risk of stockouts.

8.4 Port Dwell Time as the Dominant Variance Driver

While vessel transit accounts for 40.87 days of the journey, the 5.91-day port dwell or the Cargo Availability Gap is the critical driver of lead-time volatility. This focus on terminal nodes is supported by Ozkan & Kilic (2019), who argue that the reliability of ports (vertices) has a more dramatic impact on network success than the sea lanes (edges). For time-sensitive shipments, this nearly 6-day inactive period represents significant value decay and obsolescence risk. Shippers planning solely on a vessel's ETA will systematically underestimate their lead times; therefore, this dwell period must be explicitly budgeted into supply chain models to ensure true operational resilience.

9. Strategic Implications and Recommendations

9.1 Inventory and Buffer Management

The 95th Percentile as a Strategic Planning Baseline

Relying on the carrier's promised schedule creates a systematic shortfall that compounds into massive aggregate delay. With a mean lead time of 46.78 days, a shipper moving 1,200 containers annually faces over 10,000 days of cumulative delay, essentially requiring an extra month of safety stock just to cover the average gap. Adopting the 95th percentile (47.02 days) as the planning baseline ensures a 95% service level. This allows enough buffer to absorb extreme delivery outliers without disrupting operations.

Accounting for Port Dwell in ERP Systems

Most ERP systems use the "Vessel ETA" (the ship's arrival time) as the main milestone, but this date is misleading for actual planning. To improve accuracy, companies should change their internal systems to use a "Vessel ETA + Port Dwell" formula. This separation ensures that downstream production and retail stocking are scheduled based on actual inventory availability rather than theoretical arrival dates.

9.2 Strategic Sourcing and Contracting

Transitioning to Percentile-Based Service Level Agreements (SLAs)

Current carrier agreements typically rely on estimated transit times, which lack accountability regarding the actual percentage of on-time arrivals. To mitigate this, shippers should negotiate SLAs based on specific percentile performance, such as 90% cargo availability by Day 45. This approach shifts risk accountability toward carriers and provides a clear framework for penalties when systematic delays occur.

Tiered Logistics Response Protocols

Given the 28.11% probability of severe delay, shippers should implement automated response protocols triggered at specific decision points:

- **Day 40 (Yellow Alert):** If tracking shows the vessel has not cleared the Panama Canal, there is a high probability of a missed connection and a subsequent week-long delay. The response should be to notify distribution partners and prepare contingency demand strategies.
- **Day 45 (Red Alert):** If cargo has not cleared the New York terminal, the shipment is officially in the "severe delay" category. The response should be to activate emergency protocols, such as air freight for critical items or production rescheduling.

Maintaining a formal log of these incidents ensures financial accountability and provides the leverage needed to negotiate for improved carrier performance.

9.3 Economic Justification for Premium Routing

While 25% of sailings use fast steaming to cut the Pacific leg by about five days, many overlook these routes because of higher freight costs. However, it's important to assess the overall supply chain costs instead of focusing only on freight rates. For high-value cargo, reducing lead time by five days directly decreases safety stock needs. If the savings from lower inventory carrying costs are greater than the freight premium, then the faster route makes financial sense. For a shipper moving 1,200 containers annually, even a small net saving per unit can free up years of working capital and significantly reduce obsolescence risk, transforming premium routing into a strategic financial optimization.

10. Future Research Directions

Parallel Monte Carlo simulations of the Suez Canal and Cape of Good Hope routes could offer a comparison of reliability-cost trade-offs. This could help optimize routes based on geopolitical risk and cargo value. A sensitivity analysis could also measure which factors, like Panama Canal drought probabilities or labor disruption rates, have the biggest impact on the 95th percentile. This would identify operational improvements that provide the best return on investment.

Integrating Bayesian updating would transform the model into a live monitoring system where each leg's outcome updates the next. Instead of a static forecast, the system would recalculate delivery probabilities based on real-time progress. For instance, a delay at the Panama Canal would immediately adjust the final arrival timeline, ensuring the model reflects how disruptions compound throughout the journey. On a global scale, using this method on other major corridors like Shanghai-Rotterdam or Singapore-New York would help pinpoint systemic bottlenecks and highlight areas needing infrastructure investment.

11. Conclusion

This study demonstrates that reliability is not a shipping industry problem in isolation but a systems-level issue. When container reliability is poor, it creates ripple effects: manufacturers increase safety stock, retailers face stock-outs, consumers pay higher prices, and supply chains become less responsive to demand signals. The USD 9 billion per day held up during the 2021 Suez blockage is not an anomaly but a symptom of fragile maritime networks.

Improving reliability requires action at multiple levels: carriers must invest in faster, more predictable services; ports must improve terminal automation and labor stability; and the Panama Canal Authority must manage water resources to avoid drought-induced capacity constraints. Shippers, for their part, must demand better performance through contractual mechanisms rather than accepting marketed schedules as reality.

References

Aon. (2021). *What a stuck ship in the Suez Canal teaches us about supply chain risks*. Aon Thought Leadership.

Beacon / Seatrade Maritime. (2026, February 15). *Container shipping scheduling collapses in 2025*. Seatrade Maritime News.

DSV. (2025). *Operational updates: Delays at Panama Canal and global port congestion*. DSV Global Logistics.

Kuehne + Nagel. (2026). *Port operational updates and market analysis: February 2026*. myKN News.

Lindblad Expeditions. (2026). *What to expect crossing the Panama Canal: Operational guidelines*. World Expeditions.

Linerlytica. (2026). *Port congestion and backlog analysis: East Asia and North America*. Linerlytica Weekly.

Maersk. (2026). *North America market update: February 2026*. Maersk Ocean and Key Ports Update.

Neele-Vat Logistics. (2026). *Slow steaming in ocean freight: Advantages and operational speeds*. Neele-Vat FAQ.

Ozkan, O., & Kilic, S. (2019). *A Monte Carlo simulation for reliability estimation of logistics and supply chain networks*. IFAC-PapersOnLine, 52(13), 2080-2085.

Panama Canal Authority. (2026). Advisory to Shipping: Operational updates and transit slot availability. ACP Operations Department.

Port Authority of NY & NJ. (2025). 2024 Annual Airport & Port Traffic Statistics. Port Commerce Department.

Reuters. (2024, September 18). Port strike on US East Coast would spark supply chain glitches from outset, shipping firm says. Reuters Business.

SeaRates. (2026). *Distance and transit time tool: Global shipping routes*. SeaRates by DP World.

UNCTAD. (2024). *Review of maritime transport 2024*. United Nations Conference on Trade and Development.

World Bank. (2024). *The container port performance index (CPPI) 2023: A comparable assessment of container port efficiency*. World Bank Group.

World Shipping Council. (2025). *Operational performance review: Global liner networks*. WSC Trade Reports.

Zhang, D., et al. (2024). *Resilience analysis of maritime transportation networks: a systematic review*. Transportation Safety and Environment, 6(4).

Yue, Z., Mangan, J. (2024). A framework for understanding reliability in container shipping networks. Marit Econ Logist 26, 523–544 .

<https://doi.org/10.1057/s41278-023-00269-7>